Enterprise Local Area Network Configuration & Deployment Guide

A Technical Reference for VLAN Segmentation, DHCP Provisioning, and Router-on-a-Stick Inter-VLAN Routing

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Abstract

This paper presents a structured methodology for deploying a segmented enterprise Local Area Network (LAN) using IEEE 802.1Q VLAN technology. The architecture encompasses five discrete VLANs — Default, HR, IT, Servers, and Admin — each assigned dedicated subnets within the 192.168.0.0/16 address space. Inter-VLAN communication is facilitated through a Router-on-a-Stick topology, wherein a single physical router interface is logically subdivided into tagged subinterfaces. Centralised IP address assignment is achieved via an ISC DHCP server hosted in the Servers VLAN, with DHCP relay (ip helper-address) ensuring cross-VLAN lease distribution. Configuration procedures are detailed for Cisco IOS switches (SW1 & SW2), the ISC DHCP daemon on Linux, and a Cisco IOS router. Verification commands and a network plan table are also provided.

Keywords: VLAN, 802.1Q, Router-on-a-Stick, DHCP, Inter-VLAN Routing, Enterprise LAN, Cisco IOS, isc-dhcp-server

Table of Contents

1. Network Plan & VLAN Architecture

The network is partitioned into five VLANs to enforce traffic isolation between organisational units. Each VLAN is allocated a dedicated /24 subnet, providing up to 254 usable host addresses per segment. The gateway IP in every subnet corresponds to the router subinterface responsible for inter-VLAN forwarding.

VLAN ID	Name	Subnet	Gateway
1	Default	192.168.1.0/24	192.168.1.254
10	HR	192.168.10.0/24	192.168.10.254
20	IT	192.168.20.0/24	192.168.20.254
99	Servers	192.168.99.0/24	192.168.99.254
100	Admin	192.168.100.0/24	192.168.100.254

Table 1 — VLAN Allocation Plan

> *Note: Inter-VLAN routing is handled exclusively by the router using subinterfaces (Router-on-a-Stick). No layer-3 switching is required on SW1 or SW2.*

2. Step 01 — Switch Configuration (SW1 & SW2)

Both access-layer switches (SW1 and SW2) must be configured identically unless port assignments differ per physical topology. The configuration covers VLAN creation, access port assignment for end-user devices, and trunk port setup for uplinks to other switches and the router.

2.1 Create VLANs

VLAN entries must be defined in the switch VLAN database before any port assignment can reference them. The following commands create all four non-default VLANs:

```
vlan 10
  name HR
vlan 20
  name IT
vlan 99
  name Servers
vlan 100
  name Admin
```

2.2 Access Port Configuration (Client-Facing)

Each end-device port is set to access mode and assigned to the appropriate VLAN. Replace *gi0/X* and the trailing VLAN number to match the physical wiring plan:

```
interface gi0/X
  switchport mode access
  switchport access vlan X
```

2.3 Trunk Port Configuration (Switch-to-Switch / Switch-to-Router)

Uplink ports must carry tagged frames for all VLANs. IEEE 802.1Q encapsulation is explicitly configured before enabling trunk mode:

```
interface gi0/X
  switchport trunk encapsulation dot1q
  switchport mode trunk
```

3. Step 02 — DHCP Server Configuration

A centralised ISC DHCP server is deployed within VLAN 99 (Servers). The server must be assigned a static IP address (e.g., 192.168.99.10) within its subnet prior to service configuration. The configuration file (*/etc/dhcp/dhcpd.conf*) defines one pool per VLAN:

```
# VLAN 1 — Default
subnet 192.168.1.0 netmask 255.255.255.0 {
  range 192.168.1.100 192.168.1.200;
  option routers 192.168.1.254;
}

# VLAN 10 — HR
subnet 192.168.10.0 netmask 255.255.255.0 {
  range 192.168.10.100 192.168.10.200;
  option routers 192.168.10.254;     # fixed: was 192.168.1.200
}

# VLAN 20 — IT
subnet 192.168.20.0 netmask 255.255.255.0 {
  range 192.168.20.100 192.168.20.200;
  option routers 192.168.20.254;
}

# VLAN 99 — Servers
subnet 192.168.99.0 netmask 255.255.255.0 {
  range 192.168.99.100 192.168.99.200;  # fixed: was /32
  option routers 192.168.99.254;
}

# VLAN 100 — Admin
```

```
subnet 192.168.100.0 netmask 255.255.255.0 {
  range 192.168.100.100 192.168.100.200;
  option routers 192.168.100.254;
}
```

After saving the configuration, enable and start the DHCP daemon:

```
systemctl start  isc-dhcp-server
systemctl enable isc-dhcp-server
```

Two corrections were applied during review: (1) the VLAN 10 router option was incorrectly set to 192.168.1.200 — corrected to 192.168.10.254; (2) the VLAN 99 subnet mask was written as /32 — corrected to /24 (255.255.255.0).

4. Step 03 — Router-on-a-Stick Configuration

The physical link between the router and the uplink switch must be configured as a trunk on the switch side (see §2.3). On the router, a logical subinterface is created for each VLAN. Each subinterface receives the VLAN gateway IP address and an *ip helper-address* directive pointing to the DHCP server (192.168.99.10), enabling DHCP relay across VLAN boundaries.

```
! VLAN 1 — Default
interface gi0/0.1
  encapsulation dot1q 1
  ip address 192.168.1.254 255.255.255.0
  ip helper-address 192.168.99.10

! VLAN 10 — HR
interface gi0/0.10
  encapsulation dot1q 10
  ip address 192.168.10.254 255.255.255.0
  ip helper-address 192.168.99.10

! VLAN 20 — IT
interface gi0/0.20
  encapsulation dot1q 20
  ip address 192.168.20.254 255.255.255.0
  ip helper-address 192.168.99.10

! VLAN 99 — Servers
interface gi0/0.99
  encapsulation dot1q 99
  ip address 192.168.99.254 255.255.255.0
  ip helper-address 192.168.99.10

! VLAN 100 — Admin
interface gi0/0.100
  encapsulation dot1q 100
  ip address 192.168.100.254 255.255.255.0
  ip helper-address 192.168.99.10
```

> *The parent interface (gi0/0) must be in an 'up' state with no IP address assigned. The router automatically learns all VLAN subnets as directly connected routes upon subinterface activation.*

5. Verification Commands

The following Cisco IOS EXEC commands should be used to validate the configuration after each deployment step. These can be entered from privileged EXEC mode, prefixed with *do* when issued from a configuration sub-mode.

Command	Purpose
`do sh ip int br`	Verify IP addresses assigned to all subinterfaces
`do sh vlan`	Confirm VLAN-to-port assignments on the switch
`do sh run`	Review the complete running configuration
`do sh ip route`	Inspect the routing table on the router

Table 2 — IOS Verification Commands

6. Conclusion

This paper has outlined a complete, reproducible procedure for deploying a five-VLAN enterprise LAN. By enforcing strict layer-2 segmentation at the switch level and leveraging a single router interface for inter-VLAN forwarding, the design balances simplicity with security. Centralised DHCP provisioning through a dedicated server VLAN minimises administrative overhead while providing a single authoritative lease database.

Future enhancements may include port security on access ports, DHCP snooping, dynamic ARP inspection (DAI), and the migration from Router-on-a-Stick to a Layer-3 switch for higher-throughput inter-VLAN forwarding in growing environments.